



School of
Engineering

A PROJECT REPORT ON
“Temperature Controlled Dc Fan Using IC 741”

BACHELOR OF TECHNOLOGY IN
ELECTRONICS AND COMMUNICATION ENGINEERING

BY

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ABSTRACT

This circuit uses a thermistor to sense temperature. A thermistor is a type of resistor that changes its resistance based on temperature. In this circuit, the NTC thermistor (Negative Temperature Coefficient) has a resistance that decreases as the temperature increases.

The Op-Amp compares the voltage from the thermistor to a reference voltage set by the adjustable resistor (VR). As the temperature increases, the resistance of the thermistor decreases, which changes the voltage at the inverting input of the Op-Amp. When the voltage at the inverting input is greater than the reference voltage, the output of the Op-Amp goes low.

The output of the Op-Amp is connected to the base of the transistor. When the output of the Op-Amp is low, the transistor is off and the fan does not spin. As the temperature increases and the voltage at the inverting input of the Op-Amp increases, the output of the Op-Amp eventually goes high. This turns on the transistor, which allows current to flow to the fan and makes it spin.

The diode is used to protect the circuit from voltage spikes.

A small brushless DC fan (one used in computers) turns on to increase the air circulation. When the temperature returns normal, Fan automatically turns off. Diode 1N4007 is necessary to remove the back EMF when T1 turns off.

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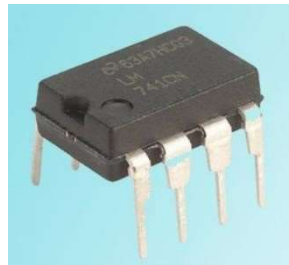
1.INTRODUCTION:

Electronic devices often generate heat during operation. Excessive heat can damage components and reduce lifespan. A temperature-controlled DC fan circuit automatically regulates fan speed based on ambient temperature, providing cooling only when necessary. This report outlines the design, working principle, and applications of a temperature-controlled DC fan using the LM741 operational amplifier (op-amp) IC.

2.COMPONENTS:

a) LM741:

The LM741 is a widely used general-purpose operational amplifier (op-amp) integrated circuit. It is known for its versatility and reliability, making it a popular choice in various Analog circuit applications. Here are some key features and characteristics of the LM741:



i. High Input Impedance:

The LM741 op-amp has a high input impedance, typically around 2 M Ω , which minimizes the current drawn from the signal source. This makes it suitable for applications where it is important to avoid loading down the input signal.

ii. Low Output Impedance

With a low output impedance, the LM741 can drive a variety of loads efficiently. This feature is important in ensuring that the output voltage is not significantly affected by the load connected to the op-amp.

iii. Wide Supply Voltage Range

The LM741 can operate over a wide range of supply voltages, from $\pm 10\text{V}$ to $\pm 22\text{V}$, making it versatile for use in different power supply

iv. Unity Gain Bandwidth

The LM741 has a unity gain bandwidth of 1 MHz, which is suitable for many general-purpose applications. This parameter indicates the frequency at which the gain of the op-amp drops to one and is a measure of the op-amp's speed.

v. Internal Frequency Compensation

The LM741 includes internal frequency compensation, which simplifies the design of stable circuits without the need for external compensation components. This makes it easier to use and implement in a variety of circuits.

vi. Slew Rate

The slew rate of the LM741 is 0.5 V/ μ s. The slew rate determines how quickly the output can change in response to a rapid change in the input signal. While not as fast as some modern op-amps, it is adequate for many applications where high-speed operation is not critical.

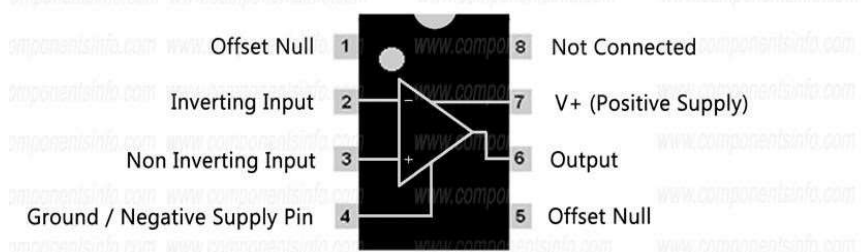
vii. Offset Voltage

The typical input offset voltage for the LM741 is around 1 mV. This low offset voltage makes the LM741 suitable for applications requiring precise voltage measurements and amplifications.

Pin Diagram:



LM741 DETAILED PIN DESCRIPTION



b) RESISTORS:

Resistors are passive electronic components that limit or control the flow of electric current in a circuit. They are crucial in setting the operating conditions of various components within a circuit, such as transistors, LEDs. Resistors primarily function to provide precise control over the voltage and current levels in a circuit. They can be used to limit current, divide voltages, bias active elements like transistors, and set gain in amplifiers.

c) POTENTIOMETER:

A potentiometer, often referred to as a "pot," is a three-terminal variable resistor that allows for manual adjustment of electrical resistance. It consists of a resistive element, typically made of carbon, with a movable contact called a wiper that slides along the resistive track. Potentiometers provide variable resistance between two of its three terminals. As the wiper moves along the resistive track, the resistance between the wiper and either end terminal changes. This property allows potentiometers to adjust the level of voltage or current in a circuit.

d) DC-FAN:

The DC fan is the actuator that provides cooling when activated by the control circuit. Typically operates at 12V or 5V, matching common power supply levels in electronic systems. Measured in cubic feet per minute (CFM), determining the cooling capacity. Designed to consume minimal power while providing adequate cooling.

e) TRANSISTOR:

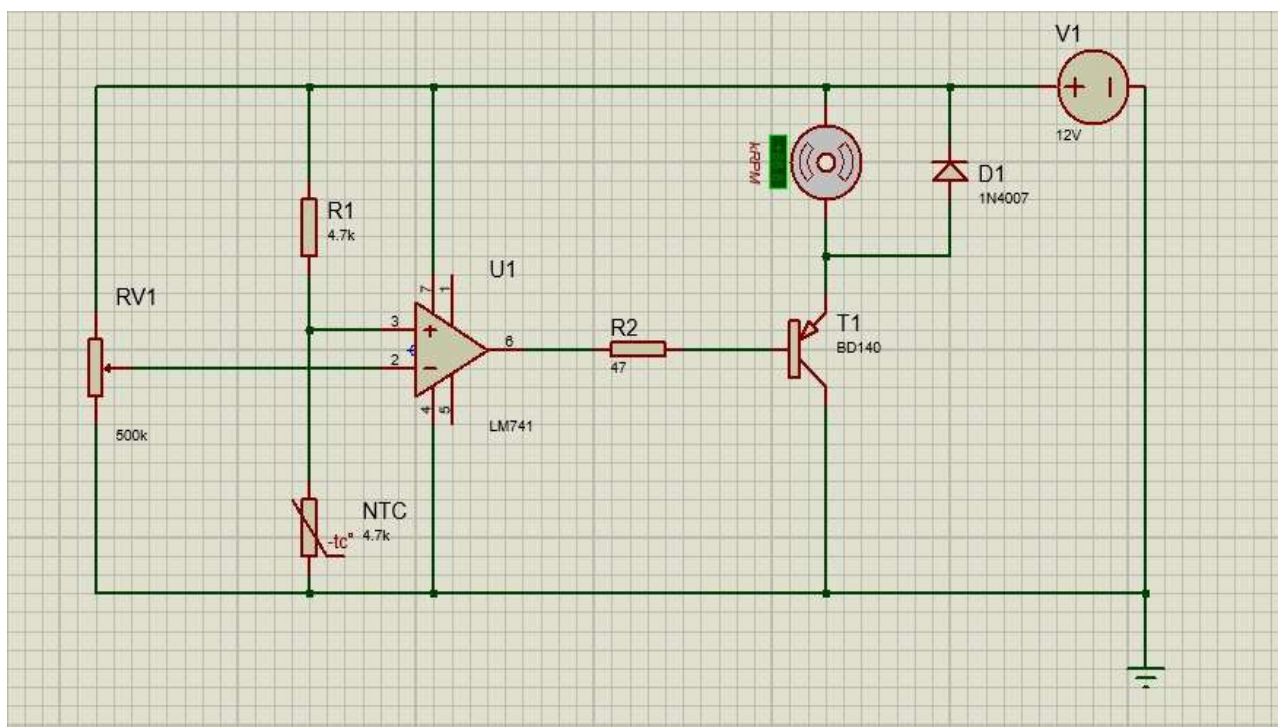
An NPN transistor is used as an electronic switch to control the DC fan based on the output from the LM741 op-amp. Amplifies the small current from the op-amp to drive the fan. Capable of fast switching, suitable for dynamic control. Low saturation voltage ensures efficient operation with minimal power loss.

f) DIODE:

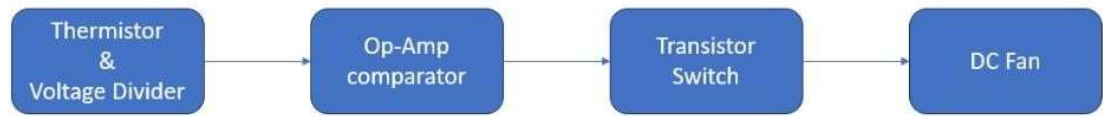
A diode (e.g., 1N4007) is used to protect the transistor from voltage spikes caused by the inductive load of the fan. Quickly responds to voltage changes, protecting the circuit. Should match or exceed the current rating of the fan. Must withstand the maximum voltage encountered in the circuit.

g) THERMISTOR:

A Negative Temperature Coefficient (NTC) thermistor is used as the temperature sensor. Resistance decreases significantly with increasing temperature. Fast response to temperature changes, providing timely control signals. Provides precise temperature measurement, essential for reliable operation.

3.CIRCUIT DIAGRAM:

4.BLOCK DIAGRAM:



5.WORKING OF CIRCUIT:

- **Thermistor:** Imagine the thermistor and the $1k\Omega$ resistor forming a voltage divider. At cooler temperatures, the thermistor's resistance is high, resulting in a higher voltage reaching the op-amp's non-inverting input (pin 3). This voltage ($V_{\text{thermistor}}$) can be approximated as:

$V_{\text{thermistor}} \approx (\text{Voltage Supply}) * (R1 / (R1 + R_{\text{thermistor}}))$ where $R1$ is the fixed resistor ($1k\Omega$) and $R_{\text{thermistor}}$ is the thermistor's resistance at that specific temperature.

- **Potentiometer:** The potentiometer sets a reference voltage (V_{ref}) at the op-amp's inverting input (pin 2). Adjusting the knob varies this voltage, allowing you to control the temperature at which the fan turns on.

Temperature Dependent Switching:

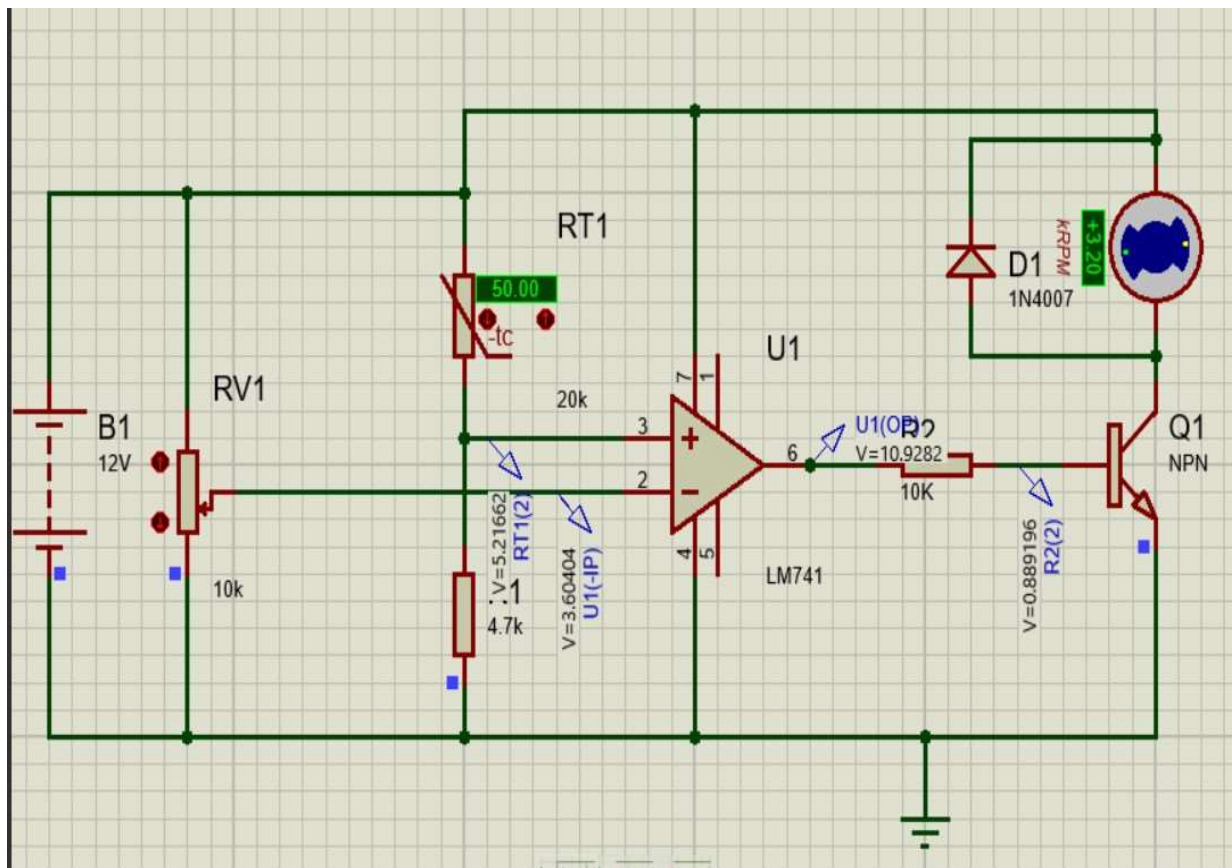
1. **Cool Temperature (Fan Off):** When the temperature is low, the thermistor's resistance is high. As a result, $V_{\text{thermistor}}$ will be higher than V_{ref} . This creates a positive voltage difference at the op-amp's inputs.
2. **Op-Amp as a Voltage Comparator:** The LM741 acts as a voltage comparator. With a positive difference ($V_{\text{thermistor}} > V_{\text{ref}}$), the op-amp saturates, driving its output (pin 6) to a high voltage (close to the supply voltage).
3. **Transistor Off (Fan Stopped):** The high output voltage from the opamp keeps the transistor in its off state (typically for NPN transistors). With minimal current flow through the transistor, the DC fan receives no power and remains stationary.

Temperature Rise and Fan Activation:

1. **Temperature Increase:** As the temperature rises, the thermistor's resistance decreases. Consequently, $V_{\text{thermistor}}$ also decreases.
2. **Approaching the Threshold:** When $V_{\text{thermistor}}$ approaches V_{ref} , the voltage difference at the op-amp's inputs gets smaller.
3. **Switching Point:** Once $V_{\text{thermistor}}$ falls below V_{ref} (due to increasing temperature), the op-amp's output voltage (pin 6) switches state and goes low (close to ground voltage).
4. **Transistor On (Fan Starts):** The low output voltage from the op-amp saturates the transistor in the opposite direction (on state for NPN). This allows current to flow through the transistor, powering the DC fan and initiating cooling.

6.OUTPUT:

➤ After a certain Threshold voltage



- The circuit regulates the DC fan's speed based on the ambient temperature detected by the thermistor.
- At cooler temperatures (thermistor resistance high), the fan remains off.
- As the temperature rises (thermistor resistance decreases), the op-amp's output voltage changes, triggering the transistor to turn on and supply current to the fan.
- The fan speed gradually increases with rising temperature, providing more cooling as needed. In summary, the circuit can be further improved by incorporating hysteresis. This introduces a positive feedback loop that creates a "dead zone" around the set temperature. Even when the temperature fluctuates slightly above or below the threshold, the fan won't constantly turn on and off. This reduces unnecessary switching and extends the lifespan of the components.

7.APPLICATIONS:

- **Computer Systems:** Cooling CPUs and GPUs to maintain optimal performance.
- **Power Supplies:** Preventing overheating in power supply units.
- **Home Appliances:** Managing temperatures in devices like refrigerators and air conditioners.
- **Industrial Equipment:** Regulating temperatures in machinery and electronic control systems.
- **Automotive Systems:** Controlling the temperature of various components to ensure efficient operation.

8. CONCLUSION:

In conclusion, the temperature-controlled DC fan using the LM741 IC is a valuable and practical project for managing thermal conditions in electronic systems. It's simple yet effective design showcases how basic electronic components can be leveraged to create a responsive and adaptive cooling system. This project not only addresses the immediate need for thermal management but also opens the door to further innovations and enhancements, ensuring that electronic devices remain efficient, reliable, and safe in their operation.